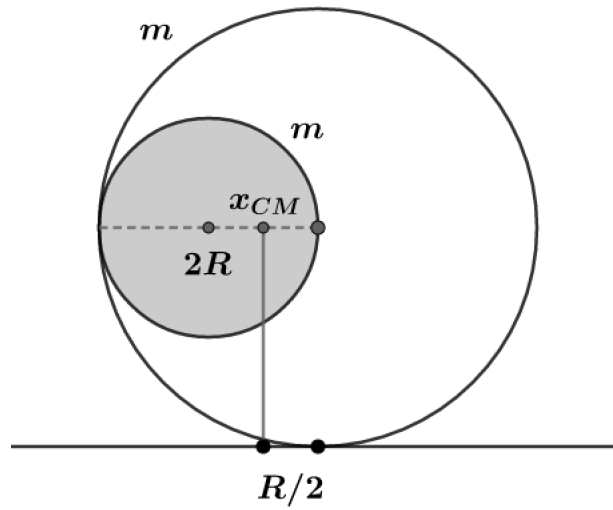


2017 F=ma Exam: Problem 3

Kevin S. Huang



Observe that there is no external force on the system in the horizontal direction, so the CM does not move in the x -direction. Since the inside mass settles to the bottom, both objects have to be located at the CM in the end. The CM is located a distance

$$x_{CM} = \frac{mR}{m+m} = \frac{R}{2}$$

to the left of the shell's initial position so the answer is B.